Md. Ridwanur Rahman

Dhaka, Bangladesh
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01711322970 Portfolio

EDUCATION

BRAC University — Dhaka, Bangladesh
BSc in Computer Science and Engineering | GPA: 3.48 — May 2022 – May 2026
Relevant Coursework: Data Structures, Algorithms, Computer Networks

Birshrestha Noor Mohammad Public College — Dhaka, Bangladesh
Higher Secondary Certificate (HSC) | GPA: 5.00/5.00 — 2021
Secondary School Certificate (SSC) | GPA: 4.83/5.00 — 2019

RESEARCH PUBLICATION

Evaluating the Impact of Adaptive CEEMDAN–Wavelet Hybrid Denoising on Foundation Model-Based EEG Detection

International Conference on Electrical and Computer Engineering (ICECE), 2026 | *Under Review*

- Developed an adaptive PCA–CEEMDAN–Wavelet hybrid denoising framework to reduce EEG artifacts and improve signal quality across multiple datasets.
- Evaluated the framework with FEMBA, achieving 73.08% accuracy, 81.74% F1-score, and 0.822 AUROC for Alzheimer's disease detection.

EXPERIENCE

EcoIntuition Academy — Remote
Data Science Intern — August 2024 – Present

- Develop and maintain EcoIntuition Academy's educational website using the MERN stack (MongoDB, React, Node.js), building responsive pages and interactive learning features.
- Design technical courses in Machine Learning, Data Science, Statistics, and Economics, integrating structured lessons, practical examples, visualizations, and assessments.
- Conduct exploratory data analysis, data cleaning, and visualization to extract insights and support data-driven learning materials.

PROJECTS

Home Management System — Live Demo | GitHub
Full-Stack Web Application | React, Vite, Node.js, Express.js, MongoDB

- Built a home and property management platform with separate functionality for property owners and tenants using React and Node.js.
- Developed RESTful APIs with Express.js and MongoDB, JWT authentication, property and tenant management, file uploads, and protected routes.
- Implemented rent and invoice management with Stripe payments, monthly invoices, payment tracking, and transaction history.

Air Quality Classification — GitHub
Machine Learning | Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Developed a four-class air quality classification system using 5,000 environmental records containing temperature, humidity, and pollutant measurements.
- Performed exploratory data analysis, correlation-based feature selection, and compared KNN, Decision Tree, and Gaussian Naive Bayes classifiers.
- Achieved 93.53% test accuracy using Gaussian Naive Bayes, outperforming Decision Tree (91.27%) and KNN (80.00%).

Uber Fare Prediction — GitHub
Machine Learning | Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Cleaned and analyzed approximately 200,000 Uber trip records, handling missing values, outliers, and feature scaling.
- Engineered temporal and geographic features, including pickup time, weekday, month, and Haversine trip distance.
- Compared Linear Regression, Random Forest, and Gradient Boosting; the best model achieved $R^2 = 0.74$ and RMSE of 4.94.

SKILLS

Languages: Python, HTML, CSS, JavaScript, C, C++, R

Frameworks & Tools: React, Node.js, Express.js, MongoDB, Git, GitHub, Scikit-learn

Technical Skills: Python GUI Development, Full-Stack Development, RESTful APIs, Data Analysis, Data Visualization, Data Cleaning, Data Management